

Introduction to Computer and Computer Science

Homework Assignment #7

Due: 11/11 0:00 am

In this homework assignment, you will practice how to apply the concept of dictionary and recursion to solve some practical issues. As you have learned all the related knowledge during the lectures, you should have the ability to accomplish this.

Problem #1: Rolling 3 dices

Please write a function called “**threeDice**” that aims to return you a dictionary with possible outcomes (the summation of these three dices) as the keys and possible combinations as its corresponding values. For example, the possible combinations of the outcome 4 are (1, 1, 2), (1, 2, 1), and (2, 1, 1). Therefore, there will be a key/value pair in your dictionary as 4: [(1, 1, 2), (1, 2, 1), (2, 1, 1)]. You can assign arbitrary number of parameters for your function (0, 1, 2, ...etc.).

`threeDice(parameters)`

The output of this function should be a dictionary with 16 key/value pairs:

The 1st key/value pair

3: [(1, 1, 1)]

The 2nd key/value pair

4: [(1, 1, 2), (1, 2, 1), (2, 1, 1)]

The 3rd key/value pair

5: [(1, 1, 3), (1, 2, 2), (1, 3, 1), (2, 1, 2), (2, 2, 1), (1, 3, 1), (3, 1, 1)]

And so on...

Problem #2: Rolling n dices

Please write a function called “**rollingDice**” that aims to return you a dictionary with possible outcomes (the summation of n dices) as the keys and possible combinations as its corresponding values. For example, the possible combinations of the outcome 3 for two dices are (1, 2) and (2, 1). Therefore, there will be a key/value pair in your dictionary as 3: [(1,2), (2,1)]. Meanwhile, the possible combinations of the outcome 4 for three dices are (1, 1, 2), (1, 2, 1), and (2, 1, 1). Therefore, there will be a key/value pair in your dictionary as 4: [(1, 1, 2), (1, 2, 1), (2, 1, 1)]. In this case, there will be only one parameter in this function:

rollingDice(**n**),

where n represents how many number of the dices you are going to roll.

For the case of $n = 2$:

The 1st key/value pair

2: [(1,1)]

The 2nd key/value pair

3: [(1,2), (2,1)]

The 3rd key/value pair

4: [(1,3), (2,2), (3,1)]

The 4th key/value pair

5: [(1,4), (2,3), (3,2), (4,1)]

And so on...

For the case of $n = 3$:

The 1st key/value pair

3: [(1, 1, 1)]

The 2nd key/value pair

4: [(1, 1, 2), (1, 2, 1), (2, 1, 1)]

The 3rd key/value pair

5: [(1, 1, 3), (1, 2, 2), (1, 3, 1), (2, 1, 2), (2, 2, 1), (1, 3, 1), (3, 1, 1)]

And so on...

For the case of $n = 5$:

The 1st key/value pair

5: [(1, 1, 1, 1, 1)]

The 2nd key/value pair

6: [(1, 1, 1, 1, 2), (1, 1, 1, 2, 1), (1, 1, 2, 1, 1), (1, 2, 1, 1, 1), (2, 1, 1, 1, 1)]

The 3rd key/value pair

7: [(1, 1, 1, 1, 3), (1, 1, 1, 2, 2), (1, 1, 2, 1, 2), (1, 2, 1, 1, 2), (2, 1, 1, 1, 2), ...
 (1, 1, 1, 3, 1), (1, 1, 2, 2, 1), (1, 2, 1, 2, 1), (2, 1, 1, 2, 1), (1, 1, 3, 1, 1), ...
 (1, 2, 2, 1, 1), (2, 1, 2, 1, 1), (1, 3, 1, 1, 1), (2, 2, 1, 1, 1), (3, 1, 1, 1, 1)]

And so on...

Hand in procedure:

As we had mentioned in the lecture, you should list all of your collaborators in your programs. Here is the template:

```
6
7 # =====
8 #
9 # Homework assignment #0
10 # Student: Your name, 你的名字
11 # Student ID: Your student ID, 你的學號
12 # Collaborators: John Doe, Jane Doe, ...
13 #
14 # =====
15 # Your code goes here...
16
17 |
```

Please save your code as a “.py” file, where the file name should follow this format:

108A_**hw#?**_**YourID**.py

For example,

108A_**hw#1**_**0180506**.py

Please be aware. **We are not going to accept any homework file with wrong file name or without signature.** Please double check the content of your file.

Once you have accomplished your works, you can upload your homework to the “New e3” system. There will be a section for uploading your homework.